JAN-S-553

25 MARCH 1948

JOINT ARMY-NAVY SPECIFICATION

STRONTIUM TARTRATE

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments and supersedes the following specification:

Navy Department
Bureau of Ordnance
51S59(Ord)
2 Dec. 1946

A. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

A–1. *Specifications.*—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

U. S. ARMY SPECIFICATIONS

50–0–1—General Specification for Ammunition, except Small Arms Ammunition.[1]

100–2—Standard Specification for Marking Shipments by Contractors.[1]

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[2]

FEDERAL SPECIFICATION

RR–S–366—Sieves; Standard, Testing.

A–2. *Other publications.*—The following publications, of the issue in effect on date of invitation for bids, form a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.[2]

INTERSTATE COMMERCE COMMISSION REGULATIONS

Regulations for Transportation of Explosives and other Dangerous Articles, etc.

B. GRADE

B–1. Strontium tartrate covered by this specification shall be of one grade only, as hereinafter specified.

C. MATERIAL AND WORKMANSHIP

C–1. The strontium tartrate shall be the tri-hydrate form, free from hygroscopic or deliquescent impurities and shall conform to the requirements of section E.

D. GENERAL REQUIREMENTS

D–1. See section E.

[1] Applicable only to Army purchases.
[2] Applicable only to Navy purchases.

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E. DETAIL REQUIREMENTS

E–1. *Color.*—The color shall be white.

E–2. *Chemical composition.*—The chemical composition of strontium tartrate shall conform to the requirements shown in table I.

TABLE I.—*Chemical composition.*

	Minimum	Maximum
	Percent	*Percent*
Strontium tartrate (anhydrous)...	81.00	85.00
Barium as Ba.....	.....	.010
Calcium as Ca.....	.....	.200
Free moisture	.....	.5
Iron as $FeCl_3$	.....	.025
Ammonium compounds	None	None

E–3. *Granulation.*—The strontium tartrate shall be of such fineness that 100 percent shall pass through a No. 60 U. S. Standard sieve and not more than 0.5 percent shall be retained on a No. 80 U. S. Standard sieve. Sieves shall conform to the requirements of Federal Specification RR–S–366.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. *Size of lot.*—Unless otherwise specified in the contract or order, a lot shall consist of material from not more than one manufacturing batch of strontium tartrate. (See par. H–3.)

F–2. *Sampling.*—Select 10 percent but in no case more than 10 of the containers comprising the lot for sampling. By means of a scoop remove from each selected container a sufficient amount of material to form a primary sample of approximately ½ pound. Mix this sample thoroughly, place a 4-ounce portion in a tightly stoppered bottle, and label each primary sample so that the container from which it was taken can be easily identified. Mix the remaining portion of the primary samples thoroughly and quarter until a composite sample of about 1 pound is obtained. Place the composite sample in a tightly-stoppered bottle and label so as to show the name of the material, manufacturer, contract or order number and number of pounds in the lot. All acceptance tests shall be made on the composite sample representative of the lot. Hold the primary samples for possible future examination should the composite sample fail to meet the requirements of this specification.

F–3. *Inspection.*—

F–3a. *Army.*—Inspection shall be made in accordance with U. S. Army Specification 50–0–1 and shall be made at the point of delivery unless otherwise specified in the contract or order.

F–3b. *Navy.*—Inspection shall be made at the point of delivery unless otherwise specified in the contract or order.

F–4. *Tests.*—Tests to determine compliance with the requirements of this specification shall be made in accordance with the following paragraphs. For Navy purchases, the tests shall be made at a Government laboratory unless otherwise specified in the contract or order.

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F–4a. *Strontium tartrate (anhydrous).*—Weigh accurately a 1.5000-gm. sample and transfer to a platinum dish. Char the material and then ignite the dish over a full flame of a Meker burner. Allow to cool and transfer to a 400-ml. beaker. Add 50 ml. of 0.5000 N hydrochloric acid. When all the carbonate has dissolved, remove any insoluble material by filtering quantitatively through a No. 41 Whatman, or equivalent, filter paper and wash thoroughly with distilled water. Titrate the combined filtrate and washings with 0.5000N potassium hydroxide, using methyl orange as the indicator. Calculate the percentage of strontium tartrate (anhydrous), as follows:

$$\text{Percentage strontium tartrate (anhydrous)} = \frac{(A \times B) - (C \times D) \times 11.785}{W}$$

where: A = normality of hydrochloric acid
B = volume of hydrochloric acid
C = normality of potassium hydroxide
D = volume of potassium hydroxide
W = weight of sample.

F–4b. *Barium.*—Ignite a 1.5000-gm. sample as described in paragraph F–4a and dissolve in 50 ml. of distilled water and 10 ml. of glacial acetic acid. Boil for 10 minutes to expel all carbon dioxide, filter quantitatively through a No. 41 Whatman, or equivalent, filter paper and wash with distilled water. Neutralize the combined filtrate and washings with ammonium hydroxide using litmus as the indicator and then make just acid with acetic acid. Heat nearly to boiling and add 10 ml. of 5-percent potassium dichromate and allow to settle overnight. Filter the solution through a tared Gooch crucible, dry, and ignite gently to constant weight, preferably in a muffle furnace. Cool the crucible and contents and weigh. Calculate the percentage of barium as Ba, as follows:

$$\text{Percentage barium (Ba)} = \frac{A \times 54.21}{W}$$

where: A = weight of barium chromate
W = weight of sample.

F–4c. *Calcium.*—Ignite a 0.800-gm. sample as described in paragraph F–4a and dissolve in 50 ml. of distilled water and a minimum amount of nitric acid. Heat to boiling and filter quantitatively through a No. 41 Whatman, or equivalent, filter paper. Wash the residue thoroughly with distilled water. Carefully evaporate the combined filtrate and washings to dryness in a covered beaker. Cool, rinse the cover and inside of beaker with distilled water and again carefully evaporate to dryness. Add 35 ml. anhydrous isoamyl alcohol and boil until the temperature remains constant at 131° C. for 3 minutes. Cool and filter through a No. 41 Whatman, or equivalent, filter paper. Wash the residue with anhydrous isoamyl alcohol. Evaporate the combined filtrate and washings carefully to dryness and remove most of the carbon by heating the beaker gently over a free flame. Cool, add 50 ml. 1:9 hydrochloric acid and heat to boiling. Filter, if necessary, and wash well with hot water. Add 10 ml. of saturated ammonium oxalate, heat to boiling and make alkaline with ammonium

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hydroxide adding 5 ml. in excess. Allow to settle and cool for 4 hours or overnight. Filter through a Gooch crucible and wash the precipitate and beaker thoroughly with distilled water. Transfer the crucible to the original beaker, cover the crucible with distilled water and add 5 ml. of sulfuric acid. Heat to boiling and titrate with standard 0.05N potassium permanganate. Calculate the percentage of calcium as Ca, as follows:

$$\text{Percentage calcium (Ca)} = \frac{A \times B \times 2.0}{W}$$

where: A = volume of potassium permanganate
B = normality of potassium permanganate
W = weight of sample.

F–4d. *Free moisture.*—Weigh accurately a 4.0 to 5.0-gram sample in a tared weighing bottle. Dry the sample in an oven maintained at 80° C. for 3 hours. Cool, weigh, and repeat the drying for 1-hour periods until constant weight is obtained. Reweigh and calculate the loss in weight as percentage of free moisture in the material.

$$\text{Percentage free moisture} = \frac{\text{loss in weight} \times 100}{\text{weight of sample}}$$

F–4e. *Iron (as $FeCl_3$).*—Ignite a 1.000-gm. sample in a platinum dish as described in paragraph F–4a and dissolve the residue in 50 ml. water to which have been added 20 drops of hydrochloric acid. Filter and wash with a weak solution of hydrochloric acid and transfer to a Nessler tube. Prepare a blank with 50 ml. of distilled water and 20 drops of hydrochloric acid. Add an equal number of drops of 10-percent potassium permanganate to both sample and blank, until a faint pink color remains, and mix thoroughly. Match the color obtained in the sample by adding, from a graduated pipette or buret, a standard ferric solution to the blank. Calculate the percentage of iron as $FeCl_3$, as follows:

$$\text{Percentage iron } (FeCl_3) = (A)\ (B) \times 0.29$$

where: A = mg. of iron per ml.
B = volume of ferric solution.

F–4f. *Ammonium compounds.*—Place a 1.0-gm. sample of the material in a 50-ml. Erlenmeyer flask and add 10 ml. of a 10-percent solution of sodium hydroxide from a pipette without allowing the solution to touch the neck of the flask. Loosely stopper the flask with a cork from which is suspended a moistened piece of red litmus paper. Warm the flask gently. The red litmus paper changing to blue indicates ammonium compounds. Report the presence or absence of ammonium compounds.

F–4g. *Granulation.*—Place approximately 100.0 gm. of sample on the specified nest of sieves, properly superimposed and assembled with a bottom pan. Cover and shake for 10 minutes by means of a single-eccentric type mechanical sieve shaker, which imparts to the sieves a rotary motion and tapping action of uniform speed of 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. Weigh the portion passed or retained by the various sieves and calculate the results to a percentage basis.

F–5. *Rejection and resubmission.*—If the lot or any portion of the lot fails to pass the inspection tests, the manufacturer shall

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have the option of having additional analyses made, without expense to the Government, to effect complete segregation of all defective portions of the lot. The manufacturer may then remove or replace defective portions of the lot and resubmit the lot for acceptance, provided that the markings on the containers are such that complete removal or replacement of defective portions of the lot can be made to the satisfaction of the Government inspector. New samples shall be taken from the entire resubmitted lot and subjected to all of the inspection tests required by this specification. If the resubmitted lot fails to pass the inspection tests, the lot shall be finally rejected.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. *Packing.*—Unless otherwise specified by the contract or order, strontium tartrate shall be delivered in airtight commercial containers. Each container shall contain approximately 200 pounds of the material. The containers shall be packed and shipped in accordance with the requirements of Interstate Commerce Commission Regulations for Transportation of Explosives, and Other Dangerous Articles, etc.

G–2. *Marking.*—Shipments shall be marked to ensure safe handling as required by Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, etc., and in addition shipments for the Army shall be marked in accordance with the requirements of U. S. Army Specification 100–2; for the Navy in accordance with the requirements of the Navy Shipment Marking Handbook. Any special marking shall be as specified in the contract or order.

H. NOTES

H–1. *Use.*—Strontium tartrate covered by this specification is intended for use in pyrotechnic compositions.

H–2. Requests, requisitions, schedules, and contracts or orders should contain the title, number, and date of the specification.

H–3. A batch is defined as that quantity of material which has been subjected to some unit chemical or physical mixing process intended to make the final product substantially uniform.

H–4. Information as to the availability of Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, etc., may be obtained from the Interstate Commerce Commission, Washington 25, D. C.

H–5. Copies of Joint Army-Navy specifications and Federal specifications (required for Army and Air Force purchases) and U. S. Army specifications may be obtained as indicated in the "Index of United States Army, Joint Army-Navy, and Federal Specifications Used by the War Department." Copies of this Index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Services within the Department of the Army, and the Department of the Air Force will obtain copies of Joint Army-Navy, U. S. Army, and Federal specifications through established departmental channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

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H–6. Copies of Joint Army-Navy specifications and Federal specifications (required for Navy purchases), Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–7. Copies of this Joint Army-Navy specification (required for Army purchases) may be obtained from the Office, Chief of Chemical Corps, Supply and Procurement Division, Army Chemical Center, Maryland.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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